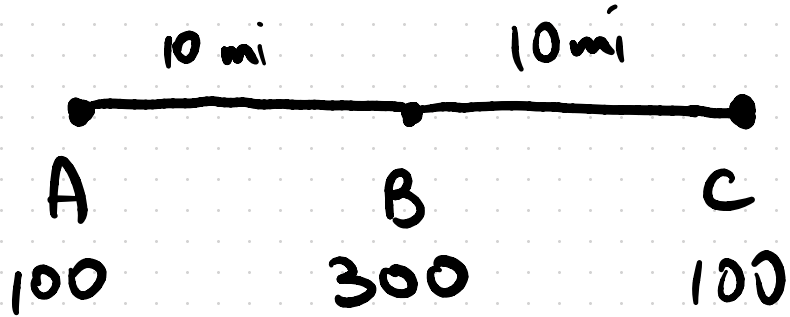
Two companies want to build a store at A, B, or C. Each customer will shop at the nearest store, and either store with 50% probability if they are equidistant. If the companies build at the same location, the rest will go up at that location and their effective number of customers decrease by 50%



The only equilibrium in this game is both play B. In any other situation, one player would benefit from changing strategies.

This is an example of a prisoner's dilemma. Both companies would benefit from cooperation, but cooperation is unstable because both companies have incentive to defect.

In a two-player zero-sum game, there is a value z such that

1) P1 can guarantee their expected payoff is at least z.

2) P2 can guarantee P1's expected payoff is at most z.

The existence of such a single value z is a consequence of linear program duality.

<u>Theorem</u>: The two linear programs

$$\begin{aligned} \text{Max} \quad & c^T x \\ \text{s.t.} \quad & Ax \le b \\ & x \ge 0 \end{aligned} \qquad\qquad \begin{aligned} \text{Min} \quad & b^T y \\ \text{s.t.} \quad & A^T y \ge c \\ & y \ge 0 \end{aligned}$$

have the same optimal value.

Many optimization problems can be seen as primal-dual LP pairs.

- Max-flow / Min-cut
- Max-matching / Min-vertex-cover
- Max-clique / Min-coloring